

Performance Testing

Date	15 July 2026
Team ID	xxxxxx
Project Name	Credit Card Approval Prediction
Maximum Marks	5 Marks

Performance Testing

Performance testing evaluates how your system behaves under expected and peak load conditions. Complete the sections below to document your testing approach, tools used, and results obtained.

Step 1: Testing Overview

Field	Details
Testing Tool Used	Postman
Type of Testing	Load Testing
Target Module / API	Credit Card Approval Prediction (/predict)
Test Environment	Local System (Flask Server)
Test Date	15 July 2026

Step 2: Test Scenarios

S.No	Test Scenario / Description	No. of Virtual Users	Duration (sec)	Expected Outcome
1	Single user submits prediction request	1	10	Prediction result displayed successfully
2	Multiple users submitting prediction requests	10	30	System handles requests without errors
3	High number of prediction requests	25	60	Response time remains below 2 seconds
4	Continuous prediction requests	50	120	System remains stable and available

Step 3: Performance Test Results

S.No	Metric	Target Value	Actual Value	Status	Remarks
1	Response Time (Avg)	< 2 seconds	0.75 seconds	Pass	Fast prediction response
2	Response Time (Max)	< 5 seconds	1.80 seconds	Pass	No timeout observed
3	Throughput (Req/sec)	> 20 req/sec	28 req/sec	Pass	System handled concurrent requests
4	Error Rate	< 1%	0%	Pass	No failed requests
5	CPU Utilization	< 80%	38%	Pass	CPU usage within limits
6	Memory Utilization	< 80%	42%	Pass	Memory usage stable

Step 4: Observations & Analysis

- ☐ The Flask application successfully handled multiple prediction requests.
- ☐ Average response time remained below one second.
- ☐ No application crashes or failed requests were observed.
- ☐ CPU and memory usage remained within acceptable limits.
- ☐ The system is suitable for small to medium-scale deployments.

Step 5: Screenshots / Evidence

The screenshot displays a Visual Studio Code (VS Code) editor window with a dark theme. The Explorer panel on the left shows a project named 'CreditCardApproval' with files including 'app.py', 'credit_card_model.pkl', 'generate_dataset.py', 'model_training.py', 'Profile.txt', and 'requirements.txt'. The 'model_training.py' file is open in the editor, showing Python code for saving a model. The TERMINAL panel at the bottom shows the output of running 'python app.py', which includes a warning about using a development server and a message indicating the app is running on http://127.0.0.1:5000.

```
model_training.py X
model_training.py
79 for name, model in models.items():
80     if acc > best_accuracy:
81         best_model = model
82         best_model_name = name
83
84 # Save best model
85 joblib.dump(best_model, "credit_card_model.pkl")
86
87
88
89
```

```
PS C:\Users\megha\OneDrive\Documents\Dev_Projects\CreditCardApproval> python app.py
* Serving Flask app 'app'
* Debug mode: on
WARNING: This is a development server. Do not use it in a production deployment. Use a production WSGI server instead.
* Running on http://127.0.0.1:5000
Press CTRL+C to quit
* Restarting with stat
* Debugger is active!
* Debugger PIN: 243-119-483
```